

ALEXANDRA GARCIA

Full Stack Software Engineer

alexgarcia1299@yahoo.com | (361) 646-3584 | in/alexandra-g1299 | alexandragarcia.me

WORK EXPERIENCE

Southwest Research Institute

San Antonio, TX (Hybrid)

Senior Software Engineer

Oct 2023 - Present

<https://its.txdot.gov/>

- Leading TxDOT's ITS (Intelligent Transportation System) platform modernization to React 18, TypeScript, ASP.NET Core 8 ahead of the 2026 World Cup, with a configurable map component powering the statewide site plus 25 district sites from one codebase.
- Consolidated district data access into a shared RTK Query and typed-config architecture spanning 6 asset domains and 13+ UI modules, cutting endpoint surface area 25% and removing duplicate-fetch and config-drift bugs.
- Architected real-time SignalR infrastructure with 60+ WebSocket hubs streaming live traffic, weather, and event data to state DOT operators.
- Built an Avigilon wrong-way vehicle detection integration in C#/.NET that converts webhook events into internal hazard alarms with image evidence and full lifecycle actions, secured with SHA-256 signing and HMAC validation.
- Delivered a Claris external-stream integration replacing polling with instant push of camera status and blackout events, cutting device-to-device communication time by ~80%.
- Leads client-facing feature demos and executes Acceptance Test Procedures (ATP) with state transportation stakeholders across Vermont, New Hampshire, and Maine; resolves critical production bugs within 48 hours and provides 24/7 on-call support to meet SLA and minimize downtime.

Software Engineer

May 2021 - Oct 2023

- Delivered end-to-end features for a multi-state traffic management platform in React, TypeScript, and C#/.NET, serving state DOT operators across multiple states.
- Spearheaded migration from a monolithic Windows application to dockerized microservices while transitioning the team from waterfall to agile.
- Established the division's NUnit/Jest testing standard across all Advance Traffic Management Software (ATMS) applications, improving production reliability.
- Mentored junior engineers, conducted technical interviews and reviewed 100+ pull requests to uphold code quality standards.

EDUCATION & CERTIFICATIONS

Bachelor of Science, Computer Science

Aug 2018 – May 2021

The University of Texas at San Antonio

Summa Cum Laude

SCRUM Master Certification

Sept 2021 – Oct 2023

Scrum Alliance

PERSONAL PROJECTS

CourtSide (NBA Game-Day Fan Hub) - Next.js 14, TypeScript, Tailwind CSS

<https://courtside-jade.vercel.app/>

- Built a full-stack Next.js NBA game-day app with server-side route handlers, mock data fallback during upstream throttling, and live games/standings/player search; deployed on Vercel with light/dark theming.

Flow State (Coffee Brewing Companion) - Angular, TypeScript, Tailwind CSS

<https://flowstate.alexandragarcia.me/>

- Built a single-page Angular app with a dosing/flow-state calculator, recipe library, and searchable bean knowledge base to help users brew consistently. Deployed on Cloudflare Pages and actively maintained with ongoing content and feature updates.

KEY SKILLS

Languages & Frameworks - React, Next.js, TypeScript, JavaScript, C#, .NET 8, Angular, Redux, SignalR

Backend & Data - REST APIs, WebSockets, OpenAPI/NSwag, MSSQL, ASP.NET Core, Entity Framework Core

Cloud & DevOps - AWS, Vercel, Cloudflare Pages, Docker, CI/CD, Git, Bamboo

Testing - NUnit, Jest, Integration Testing

Practices - Agile/Scrum, Microservices, Real-Time Systems, API Design, AI-Assisted Development (GitHub Copilot, Claude, ChatGPT)